Aitiome

An honest mechanistic-reasoning engine for the environmental exposome of neurodegeneration.

It recovers the real ones, resolves to the real neurons, and isn't fooled by the imposters.

aitiome.fly.dev / Built with Claude: Life Sciences

THE PROBLEM

What causes sporadic Parkinson’s and Alzheimer’s?

The large majority of Parkinson’s and Alzheimer’s is **sporadic** - not inherited. The leading suspect for the rest is the **environmental exposome**: the pesticides, metals, and industrial chemicals we’re exposed to over a lifetime. In 2024 the field’s leaders (Miller, Barouki, Samieri; *Nature Neuroscience*) called for AI that connects chemicals to disease *mechanism*.

Which chemicals actually drive neurodegeneration, and through what biological pathway? That is the open question this project takes on.

WHY IT'S HARD

Bioactive is not the same as neurotoxic

Tens of thousands of chemicals are **bioactive** - they light up assays. Almost none of them cause neurodegeneration. The entire difficulty is telling a true neurotoxicant apart from a compound that is merely bioactive.

This is the exact failure mode of activity-based screening: it flags everything that *does something*. Beat that, and you have something real. Fail it, and you have a watchlist no scientist can trust.

The whole project is built to win that one discrimination.

WHAT WE SET OUT TO BUILD

A discovery engine for novel drivers

The original pitch: an AI that scans the environmental chemical universe and surfaces *novel* candidate drivers of neurodegeneration, validated by recovering the known neurotoxicants.

Before writing code, we did a multi-round data reconnaissance to check whether that was actually possible on public data. It changed the project - and that is the interesting part.

THE HONEST PIVOT

The evidence redirected us

FINDING 1

No shippable unsupervised-discovery signal exists for this chemical class on public data. Seven independent axes tested; all coverage-killed or confounder-killed.

FINDING 2

The discriminating signal is

curated mechanism

, not assay activity. Bioactivity scores

worse than chance

against the decoys.

So we built the thing the data supports - and made the honesty the contribution.

WHAT AITIOME IS

Three co-equal goals

01

Validation that works

Recover the known neurotoxicants on the endorsed pathway; reject the adversarial decoys.

02

A signature visualization

The recovery-and-specificity reveal resolving into the vulnerable dopaminergic neurons.

03

Honest, calibrated framing

Confidence tiers on every result; the discovery limits shown, not hidden.

THE CORE DISCIPLINE

The recovery rule

```
positive ⇔ ( CTD curated Parkinson's DirectEvidence ) OR ( registered neuro-AOP stressor )
```

Curated signals are diagnostic.
Two independent curation efforts. A fixed logical rule, no fitted parameters to overfit.

Assay activity is corroboration only.
Anti-diagnostic here - the engine **never** gates a positive call on bioactivity.

THE SIGNATURE VISUAL

The endorsed cascade, grounded, resolving into the vulnerable neurons

MIE complex-I inhibition → mitochondrial dysfunction → nigrostriatal dopaminergic degeneration → parkinsonian deficits. Grounded in MitoCarta Complex-I subunits and the Kamath SOX6/AGTR1 vulnerable dopaminergic neurons.

THE RESULT, ON THE RECONNAISSANCE GROUND TRUTH

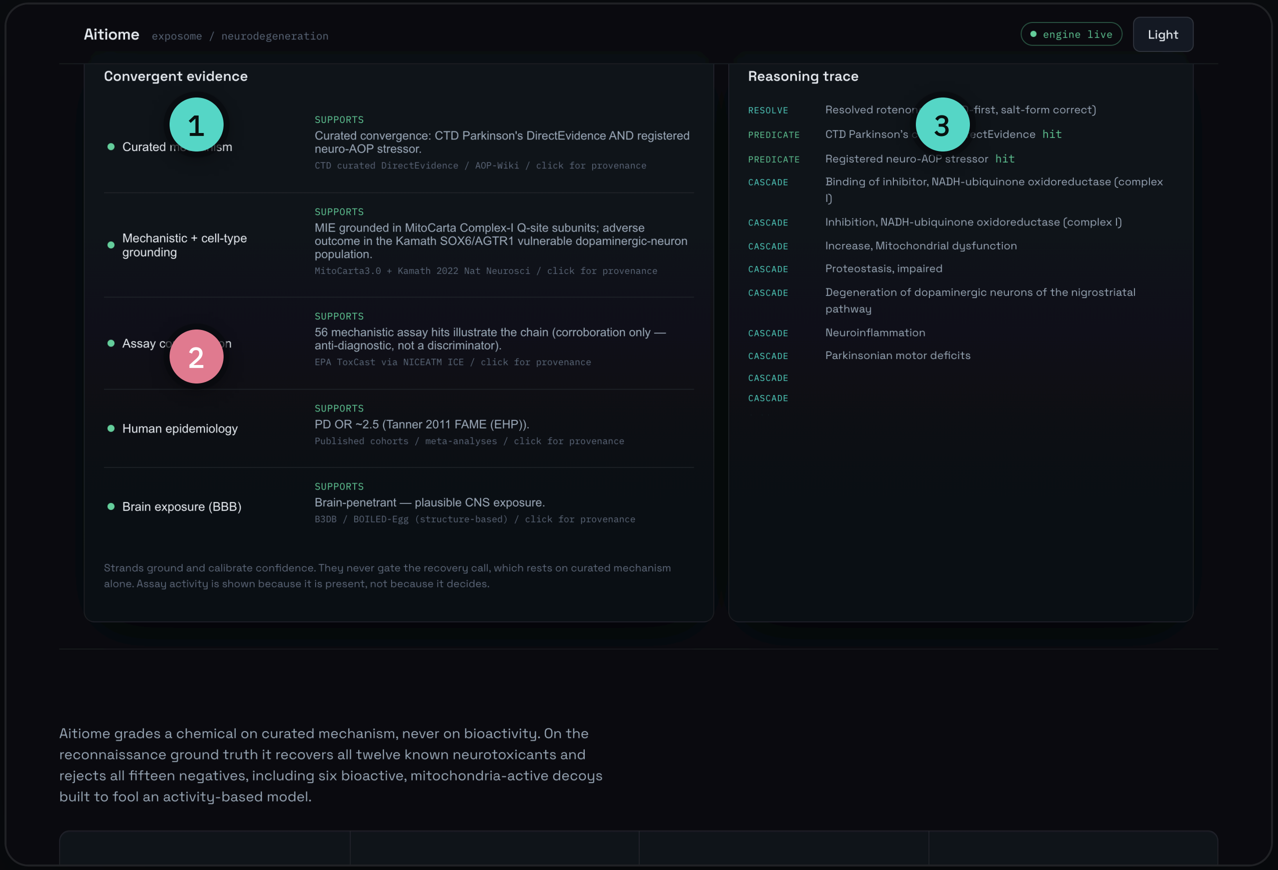
Zero errors

12/12	15/15	6/6	0
known neurotoxicants recovered	negatives correctly rejected	bioactive decoys not fooled	false positives + false negatives

12 positives (6 assay-recovered + 6 curated-anchored) - 15 negatives, including 6 mitochondria-active adversarial decoys built to fool an activity model.

THE LIVE PRODUCT - ONE RESOLVE, THE WHOLE ASSESSMENT

Convergent evidence, and the pathway reconstructed



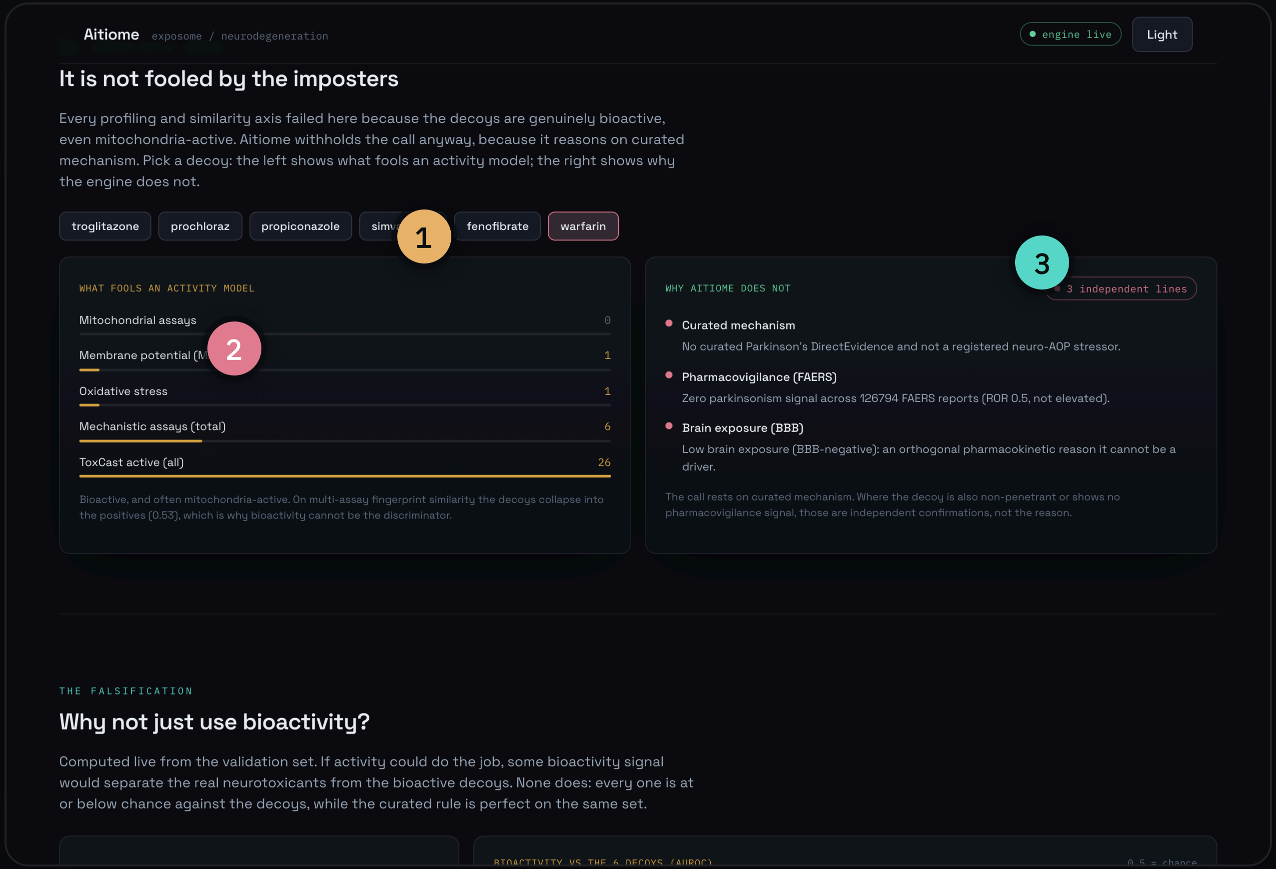
1 Five convergent evidence strands, each with provenance

2 Assay activity is corroboration only - never a discriminator

3 The endorsed AOP, reconstructed edge by edge

THE SPECIFICITY CENTERPIECE, LIVE

It is not fooled by the imposters



1 Pick any bioactive decoy

2 What lights up an activity-based model

3 Rejected on three independent lines at once

Why not just use bioactivity?

If activity could do the job, some signal would separate the 12 positives from the 6 decoys. None does - every one is at or below chance (0.5).

Curated rule on the same set: **perfect (0 errors)**. The decoys are, if anything, more bioactive than the real neurotoxicants.

ANSWERING THE SHARPEST CRITIQUE, EMPIRICALLY

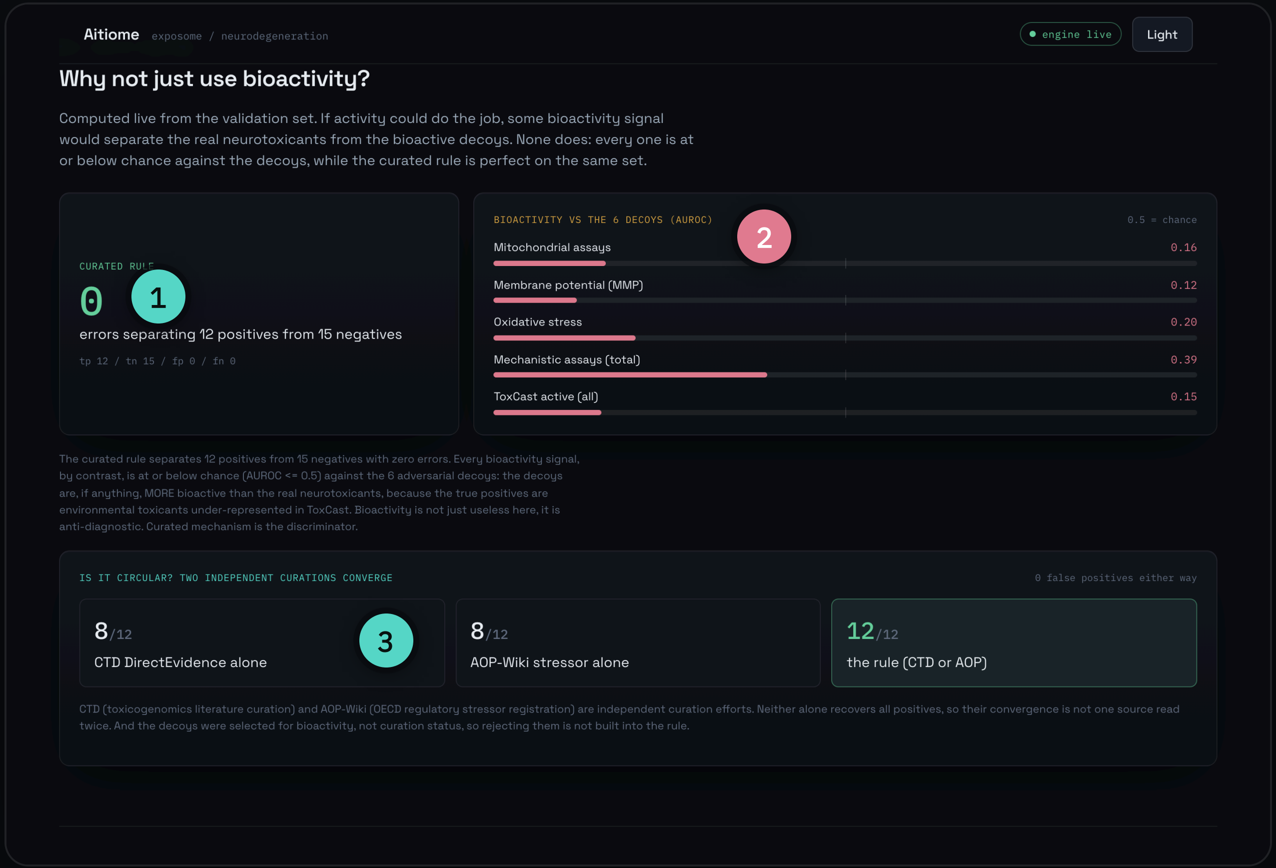
Is it circular? Two independent curations converge

8/12	+	8/12	→	12/12
CTD DirectEvidence alone		AOP-Wiki stressor alone		the rule (CTD or AOP)

CTD (toxicogenomics literature) and AOP-Wiki (OECD regulatory) are independent efforts; neither alone suffices, so this is not one source read twice. Both are **0/15** false-positives. And the decoys were selected for bioactivity, not curation status - so rejecting them is not baked in.

NOT A SLIDE - COMPUTED LIVE FROM OUR OWN DATA

The whole argument, running in the product



1 Curated rule: zero errors on the set

2 Every bioactivity signal at or below chance

3 Circularity answered: two independent curations converge

THE HONEST PART

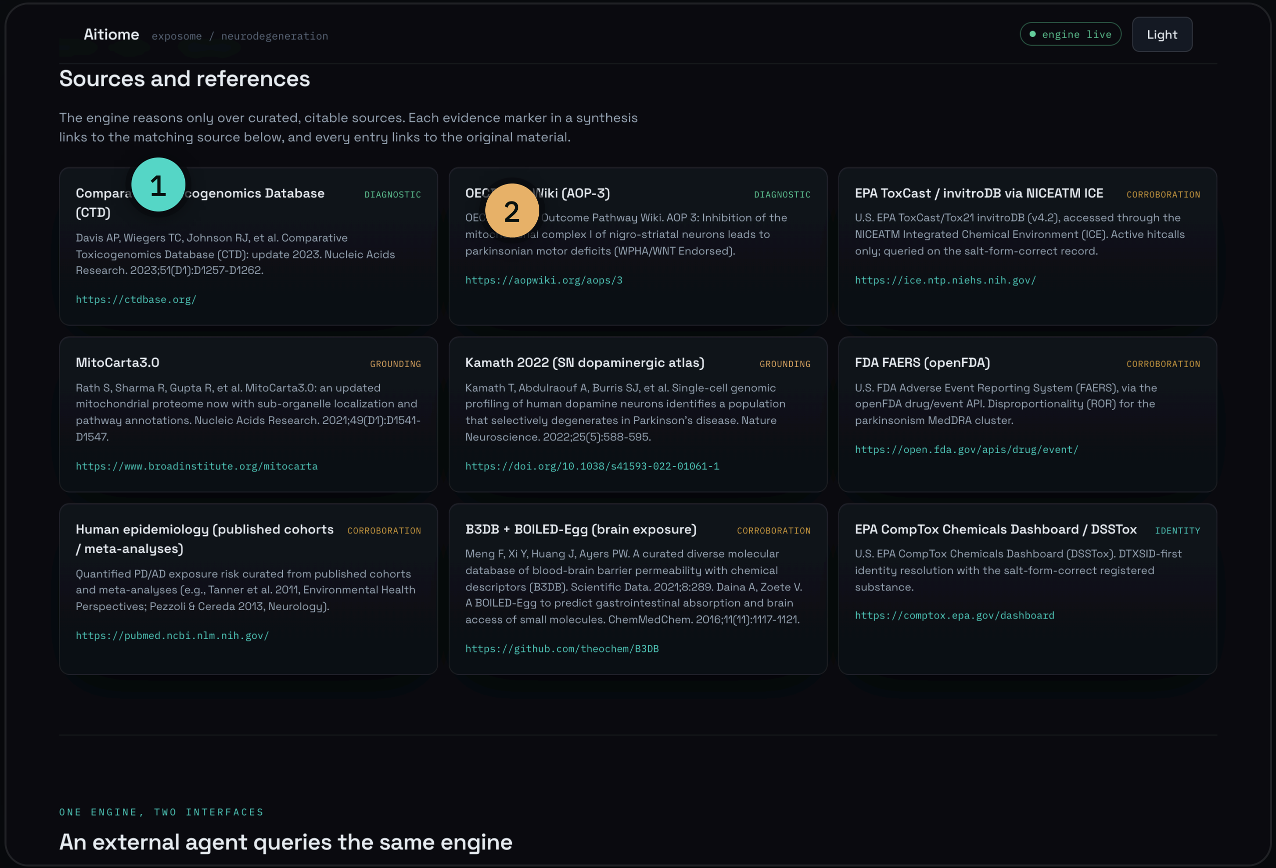
Where AI-driven discovery works, and where it does not

LINCS L1000	COVERAGE-KILLED
EPA HTTr	COVERAGE-KILLED
EPA HTPP	COVERAGE-KILLED
Structure / QSAR	CONFOUNDER-KILLED
ToxCast fingerprint	CONFOUNDER-KILLED
ComptoxAI / AlzKB	COVERAGE-KILLED
Neural-specific subset	QUALIFIED LEAD ●
Boltz-2 Q-site	CONDITIONAL LEAD ●

Seven axes coverage- or confounder-killed. Two honest, unproven leads remain (reported with explicit N). Discovery is a map, never a predictor.

AUDITABLE END TO END

Every claim links to its primary source

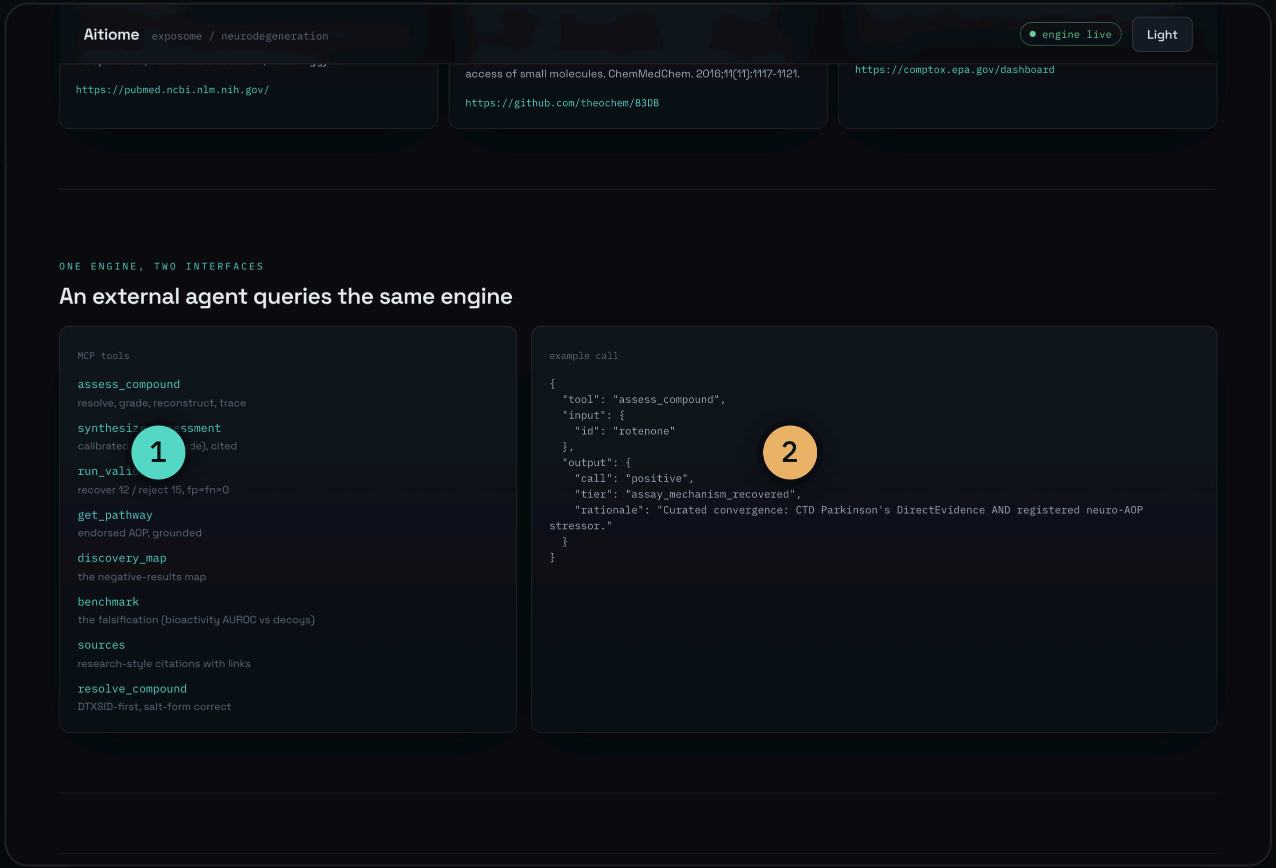


1 Nine primary sources, each a resolvable citation

2 Tagged by role: diagnostic, corroboration, grounding

ONE ENGINE, TWO INTERFACES

A scientist and an agent query the same tools



1 Eight MCP tools - the same engine, exposed to agents

2 An agent gets the same graded, cited call

BUILT WITH CLAUDE, END TO END

Claude Code builds it. The API reasons. Claude Science verifies.

Claude Code

Built the whole system - two parallel streams (engine and visualization) over a versioned contract, the deterministic core, the falsification harness, the hero, the deploy.

Claude API (Opus 4.8)

The in-app evidence-reasoner (explains, never decides) and a command-line curation agent that assembles curated evidence for a new chemical with web tools.

Claude Science

The evidence-assembly and **verification**

layer: it checks a draft against the primary sources and produces a reproducible artifact, turning a hypothesis into a curated call.

A full-stack Claude pipeline - and Claude still never enters the decision path.

Deterministic core, auditable end to end

- One Go binary embeds the curated data; the recovery decision is **deterministic** - no LLM in the decision path.
- Transport-agnostic service, thin HTTP + MCP adapters - a scientist and an external agent query the same tools.
- DTXSID-first, salt-form-correct identity (the paraquat-dichloride trap, tested).
- Every claim links to the original source (CTD, AOP-Wiki, MitoCarta, Kamath, ToxCast/ICE, FAERS, B3DB).
- **Beyond the 27:** Claude Science assembles curated evidence, `POST /assess-curated` grades it - an unverified draft is a hypothesis, never a recovery.
- ~13 MB container, live on fly.io. Reproducible: same input, same output.

WHAT WE LEARNED ALONG THE WAY

Challenges, and what they taught us

- **Novelty was the wrong optimization.** Five prior-art scans found the whole design space already published - so we built the validated prototype the field called for, honestly positioned.
- **Bioactivity is anti-diagnostic.** The deepest surprise: on our own data it scores worse than chance against the decoys. Activity cannot be the discriminator.
- **Identity correctness is day one.** The paraquat salt-form trap silently returns "no data" and corrupts everything downstream.
- **Our own test harness caught our own overclaim.** A red-team test we wrote failed, and forced the honest, stronger statement. Ground claims in executable evidence.

THE TAKEAWAYS

What this hackathon produced

A result that holds

Recovery works (12/12), specificity is proven by falsification (bioactivity at or below chance), and the discovery limits are mapped, not hidden.

Honesty as the edge

For a mechanism-first audience, a calibrated, auditable engine beats an overclaimed watchlist. Every claim is cited to its source.

Shipped, live, reproducible

Built end to end on Claude, deterministic where it must be, running at aitiome.fly.dev with an MCP interface for agents.

SAID BEFORE YOU ASK

Limitations and next steps

- A curated benchmark with an adversarial control (27), not a population generalization study.
- Recovery shares curated provenance with the labels - defended by the independent-source ablation and the orthogonal negatives; it is a sanity check, not the headline.
- Anchored on the Parkinson's / mitochondrial route; Alzheimer's is a scaffold extension.

- **Next:** a larger, blinded neural-specific reference set to power the one qualified lead (AUROC 0.72, currently perm-p 0.155).
- **Next:** run the bounded Boltz-2 Q-site benchmark - physics, not annotation or activity.
- **Next:** extend the endorsed scaffold to an Alzheimer's hallmark.

Less flashy than the original pitch.
Far more trustworthy.

Validated recovery, adversarial specificity proven by falsification, and an honest map of the limits - for a mechanism-first audience, that is the point.

aitiome.fly.dev / github.com/jonradoff/aitiome